



ELEC4848 Senior Design Project 2025-2026

Department of Electrical and Computer Engineering

[Jasminder-Singh] (3033108235)

[Multi-Modal Edge Inference for Autonomous Driving]

Supervisor: Dr. Q. Zeng

Second Examiner: Prof. Yuanwei Liu

Abstract:

This senior design project investigates the development of an optimized inference model that integrates multimodal sensor data, specifically LiDAR and monocular camera inputs, for deployment on computationally constrained edge devices. To address the inherent challenges of executing localized, lightweight artificial intelligence models, this study employs a fusion strategy combining LiDAR point clouds with image-based bounding boxes. Furthermore, model compression techniques are utilized to refine the balance between detection latency and predictive accuracy. By utilizing the KITTI dataset, the model achieves a balance between performance and accuracy. The findings underscore the viability of such models for real-time autonomous driving applications where both hardware resources and power consumption are strictly limited. However, results also indicate significant efficiency of bottlenecks when larger model variants are executed on lower-end edge devices. While the current iteration validates the feasibility of sensor fusion on the edge, future research will focus on migrating the architecture to high-performance edge computing platforms to further enhance operational efficiency and real-time throughput. Future work aims to prioritize improving pedestrian detection, refining height estimation, and retraining the model on larger and more diverse datasets to enhance generalization performance.

Acknowledgement

I would like to express my deep gratitude to Dr. Q. Zeng, my research supervisor, for their patient guidance, enthusiastic encouragement and useful critiques of this final year project.

List of Content

1. Introduction.....	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Scope	1
1.4 Objectives.....	2
1.5 Deliverables.....	2
1.6 Outline	2
2. Background	3
2.1 Calibration	3
2.2 LiDAR–Camera Transformation.....	3
2.3 Performance Metrics	5
3. Methodology	9
3.1 Overview	9
3.2 Detection Flow	9
3.3 Dataset	11
3.4 Fusion.....	12
3.5 Evaluation.....	14
4. Discussion	16
4.1 Training	16
4.2 Results	17
4.3 Shortcomings.....	18
4.4 Future Work.....	21
5. Conclusion	22
Reference:	23

Table of Figures

Figure 1: Transformation between LiDAR and camera sensor	4
Figure 2: Intersection over Union equation	6
Figure 3: All-point interpolation	7
Figure 4: 11-point interpolation	7
Figure 5: Detection Pipeline	10
Figure 6: Top View of the car	12
Figure 7: Workflow of Multimodal Fusion Method	14
Figure 8: Ground Truth Labels	19
Figure 9: Model's Predicted Values.....	20
Figure 10: Incorrect detection of height.....	20
Figure 11: Result from the Camera.....	21
Figure 12: Projected LiDAR Bounding Box	21
Figure 13: Overlapping Bounding Box.....	21
Figure 14: Fused Result	21

Lists of Table

Table 1: Directions of Camera and LiDAR	3
Table 2: Categorization of Different Difficulty	13
Table 3: Result of Evulation Script	16

1. Introduction

1.1 Background

There is an increasing adoption of electric vehicles worldwide that are integrated with advanced driver-assistance programs and autonomous driving capabilities (Gao et al. 2025). Driven by progress in computer vision and deep learning, autonomous vehicle research and development has advanced rapidly, supported by the availability of modern and accurate sensing technologies such as LiDAR (Yurtsever et al. 2020).

A critical part of the inference is the object detection model, which provides the basis for the program with a clear and comprehensive understanding of the vehicle's environment. This is critical for autonomous driving applications providing safe and precise operations that directly influence the vehicle's decision-making and its response to the environment.

1.2 Problem Statement

Despite recent advancements, existing academic reviews often focus on single sensing modalities, such as convolutional neural networks, and lack a cohesive framework summarizing multimodal fusion strategies. (Shi et al. 2025). Furthermore, a significant body of research focuses solely on more powerful edge computing device (Cui et al., 2019; Dasgupta et al., 2022; Levinson et al., 2011) because sufficient resources are provided resulting in a lack of exploration of resource-constrained edge devices and optimization of models for inferencing. In addition, there is a lack of optimization of computer vision models, which can result in latency bottlenecks (Baghbanbashi et al., 2023).

1.3 Scope

This senior year project aims to develop an optimized inference model and develop a program to run and to be deployed on a lesser powerful edge device, namely NVIDIA Jetson Nano Orin Super, while investigating quantization techniques to enhance inference throughput.

1.4 Objectives

Camera and LiDAR inference models contain tunable parameters that significantly influence detection accuracy, latency, and resource utilization. This project investigates these parameters to optimize inference performance. Additionally, quantization will be applied to explore the effectiveness of various augmentation and calibration parameters in maintaining predictive accuracy.

1.5 Deliverables

The primary deliverables are as follows:

1. A Python program will be created to run and analyze the inference results and output its results. This program will create a combination of the results of the camera and LiDAR inference to generate 3D bounding boxes around the detected objects
2. The program will be benchmarked on the Jetson Nano Orin Super, and its results will be recorded.

1.6 Outline

The outline is as follows:

Chapter 2 will introduce general terminology, providing essential knowledge to the technical aspects of the project. Chapter 3 details the methodology, including the system overview, sensor fusion strategy, and tracking algorithms. Chapter 4 discusses the training, the shortcomings and future work. Finally, Chapter 5 concludes the project, summarizing the findings and suggesting potential directions for future research.

2. Background

2.1 Calibration

Each sensor reports measurements in its own distinct coordinate system; therefore, transformation methods are required to convert data between coordinate frames and represent measurements consistently across sensors. Through calibration, a 3D point measured in the LiDAR frame can be projected into the 2D image coordinate system (pixel coordinates) and correlated with the corresponding camera geometry.

This process is referred to as extrinsic calibration and estimates the rigid-body transformation (rotation and translation) between sensors. Extrinsic calibration is essential for sensor fusion because it enables spatial alignment of measurements, providing a more reliable understanding of the environment and supporting downstream tasks such as object detection and tracking.

In addition to extrinsic calibration, the intrinsic calibration of the camera is essential parameter for the camera's internal geometry. It describes properties such as the focal length, principal point and lens distortion that are used to map 3D world coordinates accurately onto the 2D image plane

2.2 LiDAR–Camera Transformation

Figure 1 illustrates the framework involving both the camera and the LiDAR sensors used to capture the scene. Point P denotes the observed point in 3D space as seen in Figure 1. P_c and P_L represent this point in the camera and LiDAR coordinates system, respectively. Spatial transformation of these systems is achieved using the extrinsic parameters R (rotation) and t (translation). The coordinate axes for both sensors are summarized in Table 1. The system are defined as $P_C = [X_c Y_c Z_c]^T$ and $P_L = [X_L Y_L Z_L]^T$. Furthermore, point P projected onto the 2D image plane as $p_C = [u \ v]^T$.

Equation 1 defines the intrinsic parameters, which maps the 3D point P_c in the camera

coordinates to the 2D point p_c on the image plane.

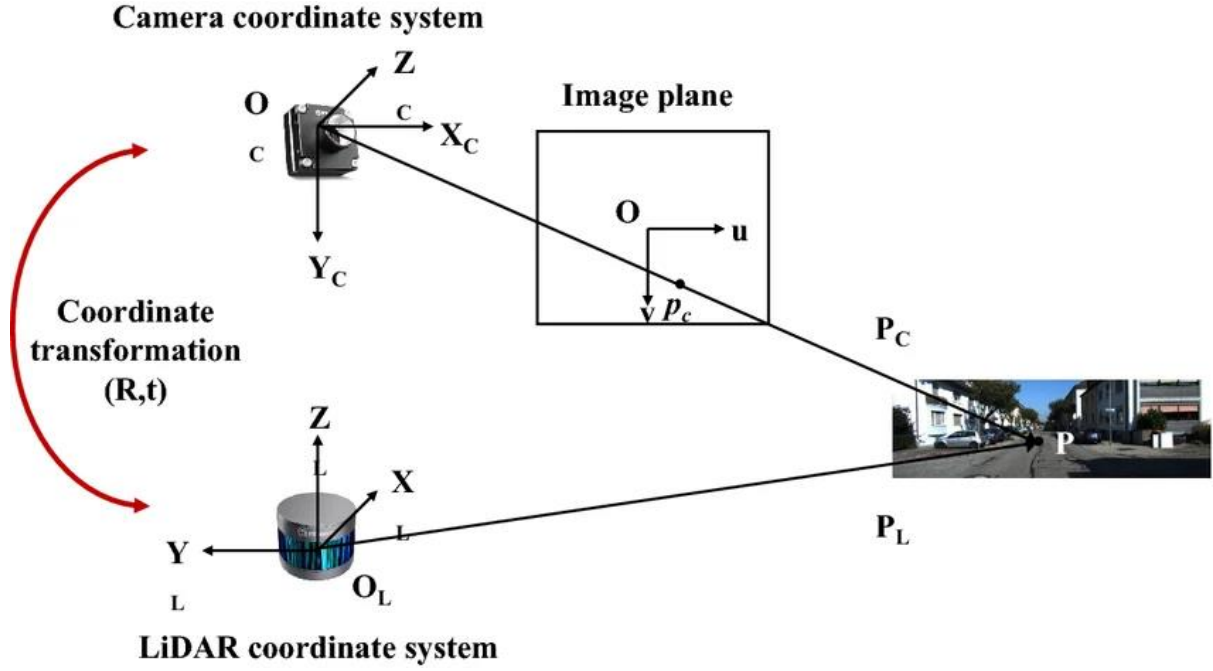


Figure 1: Transformation between LiDAR and camera sensor

$$Z_C \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_C \\ Y_C \\ Z_C \end{bmatrix} = K_C \begin{bmatrix} X_C \\ Y_C \\ Z_C \end{bmatrix}$$

Equation 1

Sensor	X-axis	Y-axis	Z-axis
Camera	Right	Down	Forward
LiDAR	Forward	Left	Up

Table 1: Directions of Camera and LiDAR

In Equation 1, Z_C represents the depth scale factor, while the focal lengths are represented as f_x and f_y (in pixels). The variable (u_0, v_0) represent the principal point, and s denotes the skew coefficient, which is zero if the image axe is perpendicular. The matrix K_C is the intrinsic projection matrix.

The extrinsic parameters used to transform 3D LiDAR points P_L into the camera coordinate P_C are defined in equation 2.

$$\begin{bmatrix} X_C \\ Y_C \\ Z_C \end{bmatrix} = \mathbf{R} \begin{bmatrix} X_L \\ Y_L \\ Z_L \end{bmatrix} + \mathbf{t} = [\mathbf{R} \ \mathbf{t}] \begin{bmatrix} X_L \\ Y_L \\ Z_L \\ 1 \end{bmatrix}$$

Equation 2

In equation 2, T represents the translation vector, and R stands for the rotation matrix.

Combining equation 1 and 2 yields the total transformation between the 3D LiDAR points and the 2D image plane point as represented in equation 3. This enables the mapping of the image coordinates to LiDAR coordinates and vice versa.

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}_c [\mathbf{R} \ \mathbf{t}] \begin{bmatrix} X_L \\ Y_L \\ Z_L \\ 1 \end{bmatrix}$$

Equation 3

2.3 Performance Metrics

To determine the effectiveness of the object detection system, quantitative performance metrics are required to ensure the system is accurate and reliable. These metrics utilize the outputs of object detectors: the object label, bounding box and the confidence score.

Primary metrics are used to evaluate a model performance (Geiger et al. 2012):

- **Ground Truth (GT):** Manually labelled bounding boxes and object classes within the dataset. This data is used to train and evaluate the model by comparing the prediction of the model against the ground truth annotation
- **True Positive (TP):** A correct prediction where the model correctly identifies the object class and the bounding box overlaps significantly with the GT.
- **False Positive (FP):** An incorrect prediction where the model detects an object that does not exist in the GT or significantly misplaces the detection.
- **False Negative (FN):** A missed detection where an object is present in the GT data but is not identified by the model.

Intersection over Union (IoU):

IoU measures the overlap between the GT bounding box (BB_{GT}) and the predicted bounding box (BB_P). It is the ratio of the area of intersection to the area of union as seen in Figure 2 (Shah, 2023).

$$IoU = \frac{area(BB_{GT} \cap BB_P)}{area(BB_{GT} \cup BB_P)}$$

Equation 4

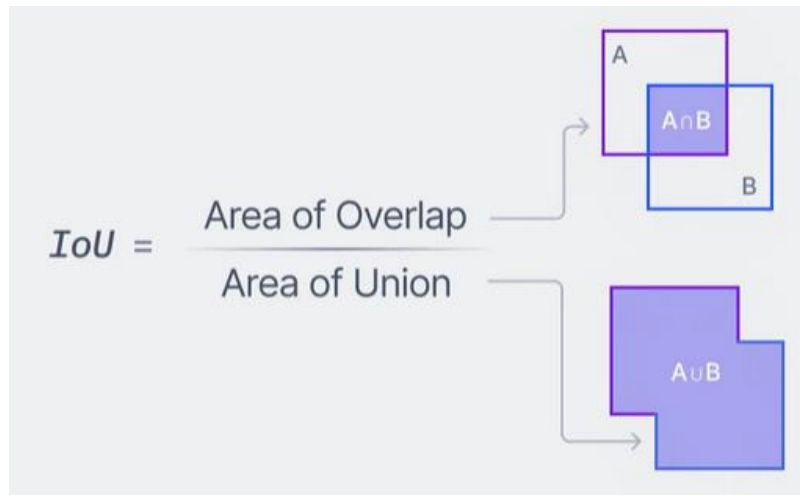


Figure 2: Intersection over Union equation

Precision:

Precision, as defined in Equation 5, measures the proportion of correct positive detections relative to all positive predictions made by the model.

$$precision = \frac{TP}{TP + FP}$$

Equation 5

Recall :

Recall, as defined in Equation 6, measures the proportion of actual ground truth objects that

were correctly identified. It serves as a measure of the model's completeness.

$$recall = \frac{TP}{TP + FN}$$

Equation 6

Average Precision (AP) :

Average Precision (AP) evaluates the performance of the detector model for a specific category. The area under the Precision–Recall (PR) represents the model’s ability to balance precision and recall across various confidence thresholds.

AP is typically computed using all-points interpolation (Figure 3) or 11-point interpolation (Figure 4) to smooth the curves. All-points interpolation captures the full area under the PR curve by considering every unique recall level where precision changes. In contrast, 11-point interpolation averages the maximum precision values at eleven equally spaced recall levels (0.0 to 1.0) (Hui, 2018).

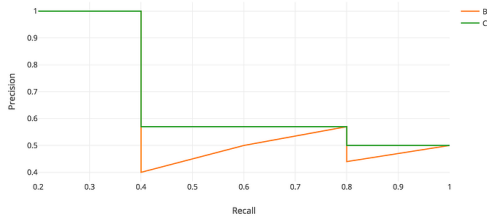


Figure 3: All-point interpolation

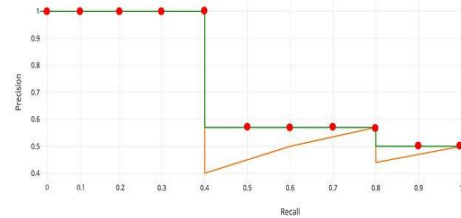


Figure 4: 11-point interpolation

Mean Average Precision (mAP):

The Mean Average Precision (mAP) is the arithmetic mean of AP values computed across all unique object categories within the dataset. It provides a single summarized metric for the overall accuracy of the model where C is the number of classes and AP_c is the AP for class c :

$$mAP = \frac{1}{C} \sum_{c=1}^C AP_c$$

Equation 7

3. Methodology

3.1 Overview

The methodology section outlines the structured approach taken to design, implement, and evaluate the multimodal sensor fusion pipeline. The primary objective is to develop a computationally efficient framework that leverages the complementary strengths of LiDAR and monocular camera data for real-time object detection on edge hardware.

3.2 Detection Flow

Figure 5 illustrates the simplified architecture of the detection pipeline, which is categorized into three primary modules: the camera subsystem, the LiDAR subsystem, and the fusion module. The camera subsystem processes visual imagery of the scene, while the LiDAR component operates on a Bird's-Eye View (BEV) representation of the captured point cloud data.

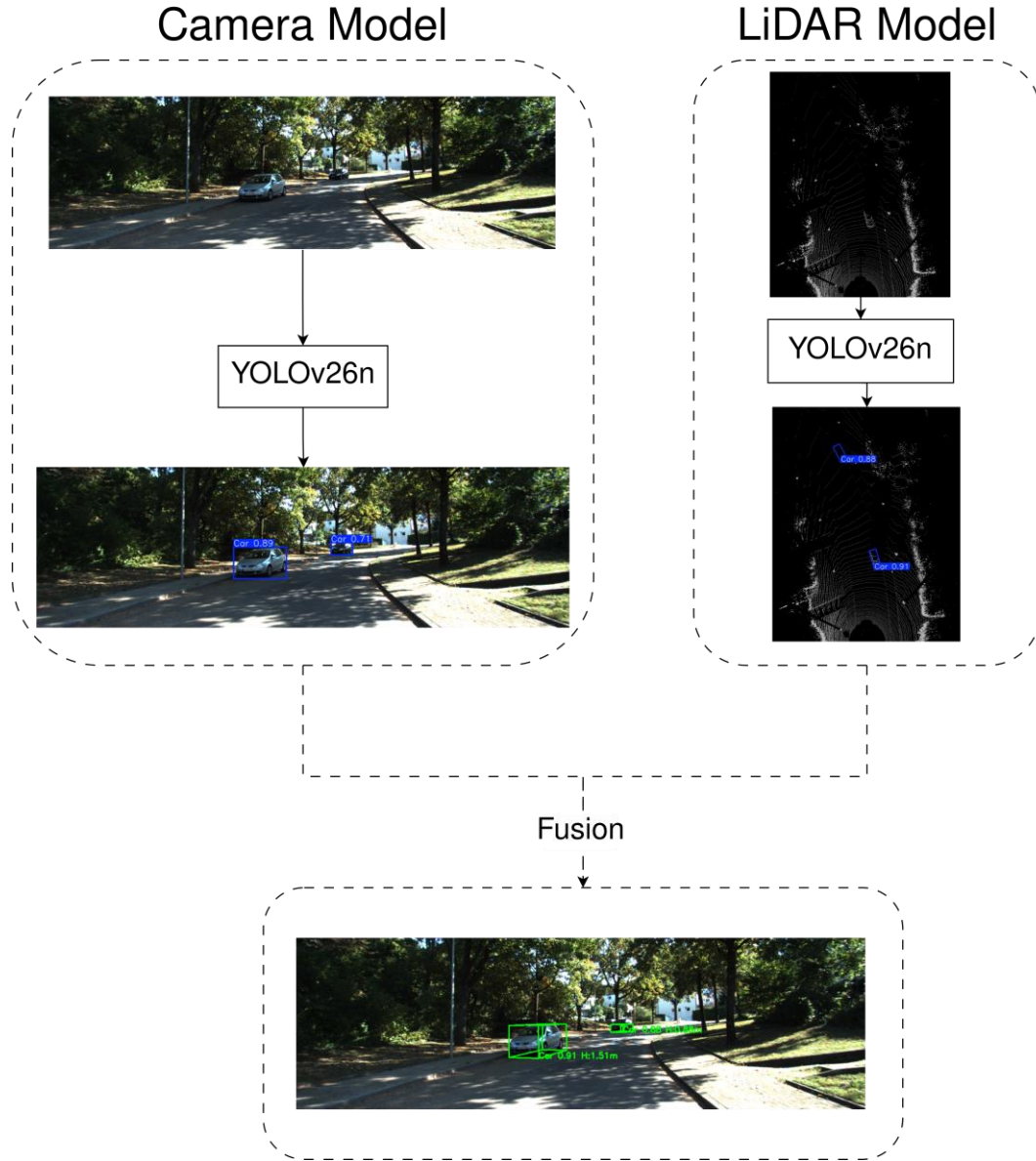


Figure 5: Detection Pipeline

As illustrated in Figure 5, the detection pipeline begins with the camera model, where the input image is processed via a YOLOv26 architecture to extract two-dimensional (2D) bounding boxes and associated confidence scores. Concurrently, the LiDAR model represents the point cloud in BEV format, which is then processed by a YOLOv26n-obd model to generate oriented bounding boxes. Subsequently, the system utilizes extrinsic calibration parameters to transform these LiDAR-space coordinates into the camera coordinate system,

enabling the projection of 3D bounding boxes onto the 2D image plane.

3.3 Dataset

The KITTI 3D Object Detection dataset (Geiger et al. 2012) was selected for this project because it provides a standardized, high-quality benchmark essential for evaluating multi-modal fusion algorithms. As one of the pioneering publicly available datasets for autonomous driving, it offers a synchronized and well-documented collection of LiDAR and camera data, allowing for direct performance comparisons with existing state-of-the-art models.. The dataset provides 7481 training images, 7518 test images, along with the corresponding LiDAR point clouds, and the calibration files. The training folder consists of manually labelled annotation, which can be projected onto the 2D image plane using the provided calibration data.

Figure 6 illustrates the sensor configuration from a top-down perspective. The setup includes four cameras, two grayscale (Cameras 0 and 1) and two color (Cameras 2 and 3), that are used for images and videos capture, alongside a Velodyne laser scanner (LiDAR) and a GPS/IMU system. The Camera and LiDAR both captures data at a frequency of 10 frames per second (FPS). The accompanying calibration files enable the transformation of coordinates between the cameras, LiDAR, and the GPS.

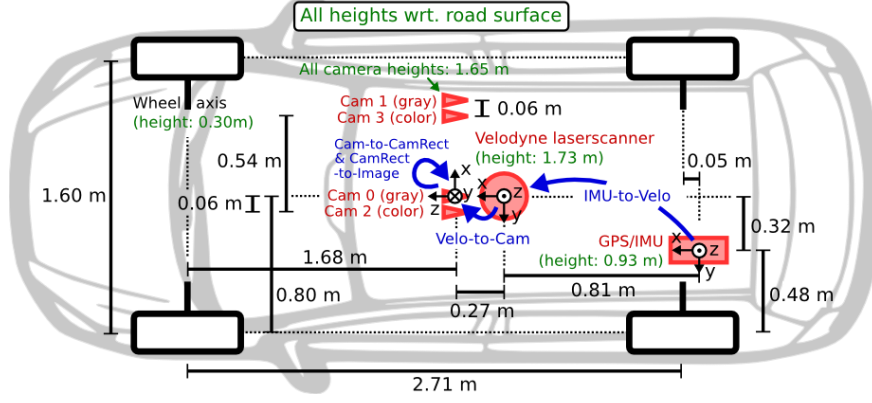


Figure 6: Top View of the car

For this project, the training and testing phases utilize data exclusively from the left-facing color camera (Camera 2) and the LiDAR sensor. While the KITTI dataset provides synchronized color imagery from both the left (Camera 2) and right (Camera 3) perspectives, the left-facing dataset was selected for this study. However, the methodology and model remain compatible with either camera source.

3.4 Fusion

Following the LiDAR inference, the initial task involves extracting the vertices of the oriented bounding boxes. These vertices serve as critical anchor points for projecting the spatial data onto the 2D image plane. Once obtained, the LiDAR coordinates are transformed into the camera coordinate system utilizing the projection matrix defined in Equation 8.

To establish the bounding box in the image plane, the maximum and minimum horizontal (x) coordinates are extracted from the projected points, while the largest LiDAR value from the corresponding bounding box is used to define the height.

Following the coordinate transformation, a validation step is performed by comparing the camera-based inference results with the projected 2D LiDAR data using the IoU metric. Based on empirical testing, a minimum IoU threshold of 30% was established to correlate the

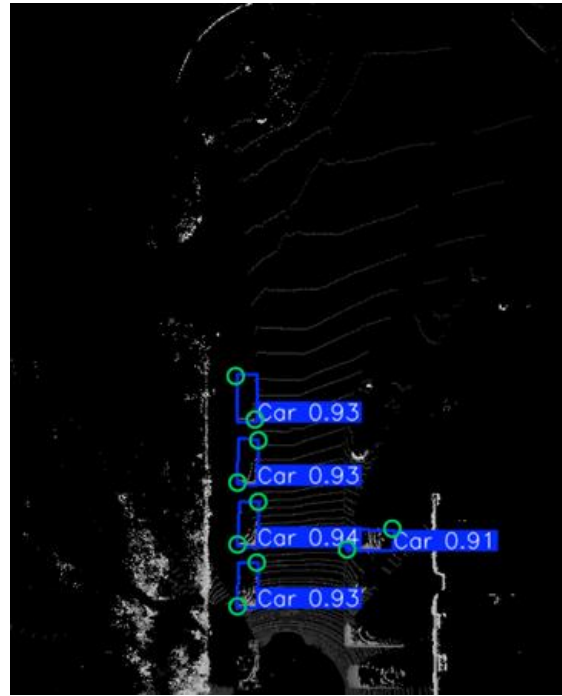
two sensor inputs. This specific threshold was selected to account for spatial variance and potential projection inaccuracies inherent in the LiDAR-derived 2D bounding boxes. If this threshold is met, the 3D bounding box is projected onto the image plane. Finally, visual construction is achieved by mapping the LiDAR corner coordinates to both the object height and the ground plane. The sequential stages of this fusion process and the resulting visualizations are detailed in Figure 7.

$$y = P(i) @ R(0) @ T_{velo_cam} @ x$$

P(i)	Camera-Image projection matrix for I'th camera
R(0)	Rectification matrix
T _{velo-cam}	Velodyne-Camera transformation matrix
@	Matrix multiplication operation

Equation 8 (Sharma, 2025)

Step 1: Retrieve the corners of the oriented bounding box and identify the maximum and minimum x-values (indicated in green).



Step 2: Project the 3D LiDAR points to generate a 2D bounding box within the image plane.



Step 3: Calculate the IoU by overlapping the camera-based and LiDAR-based bounding boxes.	
Step 4: Filter detections for an IoU > 30% and project the final 3D bounding box onto the scene.	

Figure 7: Workflow of Multimodal Fusion Method

3.5 Evaluation

To evaluate the model's performance, this project utilizes the KITTI evaluation script (Guindel, 2019). This script is built upon the standard KITTI development kit but offers enhanced documentation and streamlined processing. For every processed frame, the system generates a text file containing the following attributes for each detected object:

- **Object Class:** Identifies the category of the detected entity. The evaluation script focuses exclusively on Car, Pedestrian, and Cyclist. Other categories are excluded from the final evaluation of metrics to maintain evaluation consistency.
- **2D Bounding Box:** Defined by the 2D pixel coordinates of the top-left and bottom-right corners. This box represents for the projection of the 3D bounding box onto the 2D image plane.
- **3D Bounding Box Dimensions:** Specifies the object's physical height, width, and length, measured in meters.
- **3D Bounding Box Ground Center:** The precise spatial center of the object's base, expressed in LiDAR coordinates.
- **Confidence Score:** A floating-point value representing the model's certainty. Higher values indicate a greater probability that the object belongs to the predicted class.

The KITTI benchmark categorizes objects into three difficulty levels based on their visibility

and size within the image. Table 2 outlines the criteria used for these levels. The minimum bounding box height refers to the projected 2D height, which refers to the object's proximity to the sensors. The maximum occlusion level quantifies the entity's visibility, accounting for instances where the object is partially or significantly obscured by environmental features or other vehicles.

Difficulty	Min. bounding box height	Max. occlusion level
Easy	40 Px	Fully Visible
Moderate	25 Px	Partly Occluded
Hard	25 Px	Difficult to see

Table 2: Categorization of Different Difficulty

4. Discussion

4.1 Training

Various experimental iterations were conducted to evaluate the impact of hyperparameter configurations on model performance. These variables included the number of training epochs, data augmentation techniques, class selection, architectural scale, and input image resolution. Performance metrics were systematically recorded for each configuration to identify optimal training parameters. Initial model training was executed on an NVIDIA GeForce RTX 3050 Ti GPU. Following the training phase, the models were converted into optimized TensorRT engine files. This optimization process was essential to minimize inference latency during deployment on the NVIDIA Jetson Orin Nano Super Developer Kit.

In the initial development phase, the detection classes comprised of eight categories: Car, Van, Truck, Pedestrian, Cyclist, Person sitting, Tram, and Miscellaneous. Following preliminary training iterations, the classes were consolidated to improve model convergence and relevance. Specifically, the "Person sitting" category was merged into the "Pedestrian" class while "Miscellaneous" was excluded. The final configuration consists of six primary classes: Car, Van, Truck, Pedestrian, Cyclist, and Tram.

Hardware limitations necessitated significant adjustments during the deployment phase. The Jetson Orin Nano was unable to sustain the computational requirements of larger model architectures and high-resolution image inputs, resulting in frequent overcurrent events. These issues were attributed to the intensive power demand during inference, which triggered thermal throttling and a subsequent degradation in processing speed. To ensure system stability, the "Nano" variant of the model was adopted. Efficiency was further enhanced by converting LiDAR's input data from three color channels to a single grayscale channel. By reducing the input dimensionality, the model minimizes the total operations per inference,

thereby lowering the overall computational cost.

Under optimized conditions, the system utilizes the YOLOv26 Nano architecture to achieve a processing rate of 9.8 frames per second (fps). This performance closely aligns with the 10-fps output frequency of the camera and LiDAR sensors, ensuring near real-time synchronization for multi-modal data and fusion.

4.2 Results

Table 3 summarizes the performance results obtained from the evaluation script. The object detection performance across the three difficulty tiers, measured by Average Precision (AP) scores, generally correlates with the environmental complexity of the scene.

The car detection task achieved its highest performance in the "Easy" category, with consistent results observed across the "Moderate" and "Hard" categories. In contrast, pedestrian detection proved to be the most challenging for the model. It yielded an AP of 0.7422 in the "Easy" tier, with notable performance degradation as the difficulty level increased. Conversely, cyclist detection achieved the highest overall scores and maintained robust performance across all categories. These findings indicate that while the model is highly effective at identifying vehicles and cyclists, it faces greater difficulty accurately detecting pedestrians in complex scenarios.

Category	Easy	Moderate	Hard
Car Detection	0.8578	0.7201	0.7265
Pedestrian Detection	0.7422	0.6776	0.6346
Cyclist Detection	0.8509	0.8307	0.8067

Table 3: Result of Evulation Script

4.3 Shortcomings

A discrepancy was observed between the LiDAR-based detections and the established ground truth. This variance often stems from the labeling methodology because LiDAR training labels are generated by projecting camera-based annotations into the LiDAR domain, the model frequently detects objects that exist in the physical space but were not captured by the initial camera-derived labels. Furthermore, as demonstrated in the comparison between ground-truth labels in Figure 8 and model predictions in Figure 9, the LiDAR system encountered difficulties in identifying pedestrians. This challenge is likely a result of the sparse and ambiguous spatial representation of pedestrians in point cloud data. Such objects often exhibit geometric characteristics similar to those of the surrounding environment, complicating the distinction between the target and the background.

Furthermore, after extracting the required points, the LiDAR measurements are projected onto the camera image for visualization. However, several inconsistencies are observed. Because object height is estimated using the maximum LiDAR point height within the associated region, the resulting height can be unreliable partially when LiDAR returns are sparse or the object is partially occluded. This often produce misaligned or incorrectly scaled

3D bounding boxes, as illustrated in Figure 10.

Furthermore, a mismatch between camera and LiDAR detections can prevent successful fusion. As shown in Figure 11, the camera detector typically more accurately captures the full extent of an object, whereas in Figure 12 shows that LiDAR-based detection may have incorrect 2D projection. This partial overlap reduces the IoU between two modalities as shown in Figure 13, which can cause the fusion step to fail and prevents the resulting 3D bounding box from being accurately projected onto the camera image as seen in Figure 14.

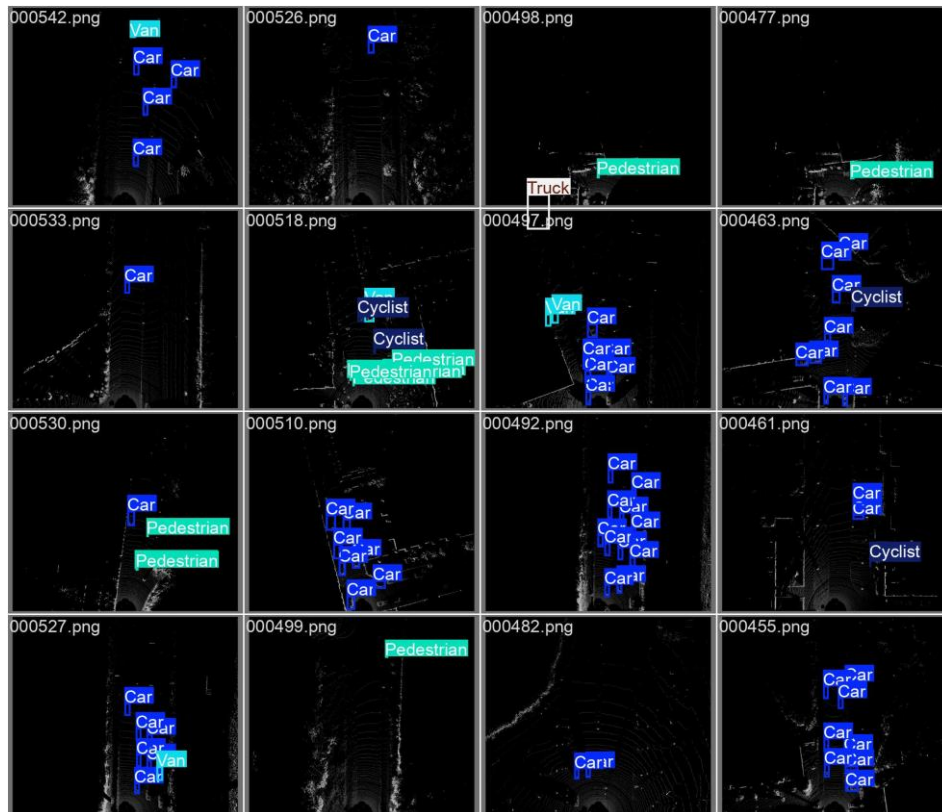


Figure 8: Ground Truth Labels

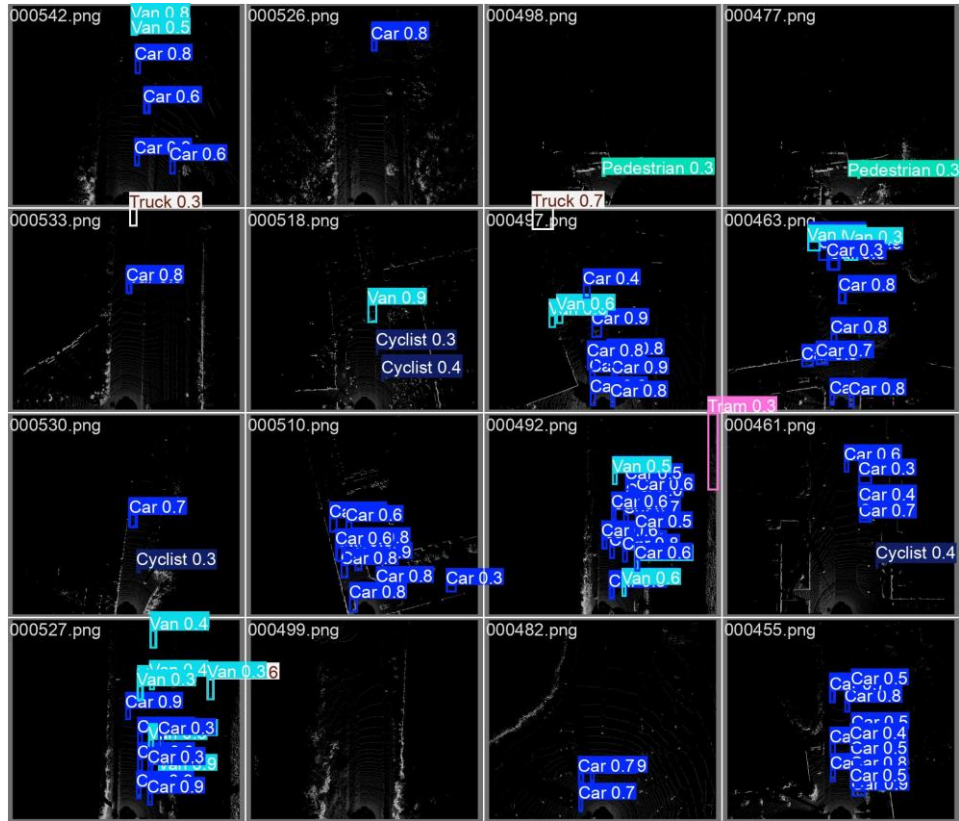


Figure 9: Model's Predicted Values



Figure 10: Incorrect detection of height



Figure 11: Result from the Camera



Figure 12: Projected LiDAR Bounding Box



Figure 13: Overlapping Bounding Box

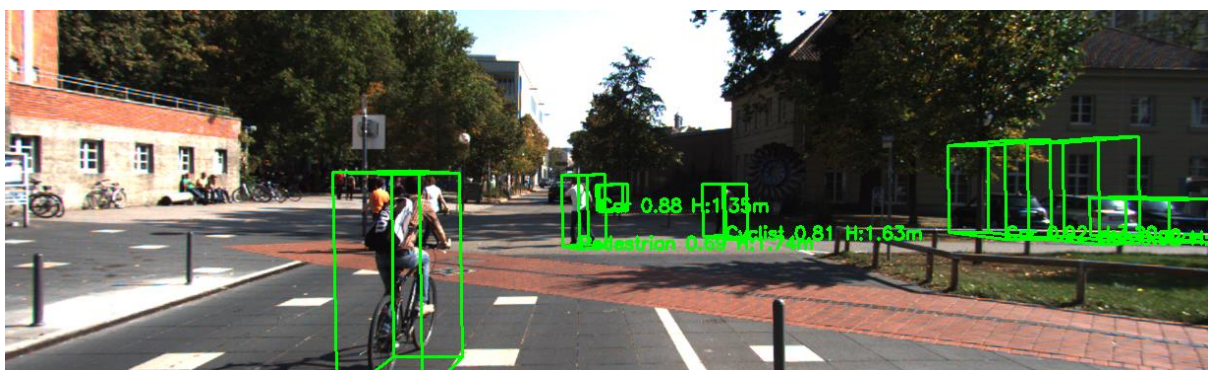


Figure 14: Fused Result

4.4 Future Work

Analysis of the current findings suggests several pathways for enhancing 3D bounding box accuracy. A primary technical challenge throughout development was reconciling high-precision detection with the limited computational and budgetary constraints of the target edge device. To maintain real-time inference within these hardware limitations, the YOLO-Nano architecture was prioritized.

Future iterations should specifically address refinements in height mapping and pedestrian detection. Furthermore, the model's predictive power is currently constrained by the scale of the training data. While the present study utilized approximately 7,000 images, benchmarking against industry-standard datasets such as Waymo (Waymo LLC, 2019) which exceeds one million samples, indicates that significant gains in generalization and performance could be achieved through large-scale data augmentation and training.

5. Conclusion

The project successfully implemented a multimodal detection pipeline by integrating camera-based YOLOv26 inference with LiDAR-based processing through coordinate transformation. The proposed model achieves an effective balance between inference performance and the hardware constraints of the NVIDIA Jetson Orin Nano. These results suggest that increasing the computational budget or utilizing more powerful GPUs would yield even higher accuracy and efficiency.

In summary, the developed model establishes a robust framework for computationally efficient 3D object detection on edge devices. To build upon these findings, subsequent research should prioritize the refinement of the detection pipeline to improve spatial precision. Furthermore, retraining the model on more expansive and diverse datasets will be essential to enhance its reliability and performance in complex, real-world autonomous driving environments.

Reference:

- Baghbanbashi, M., Raji, M., & Ghavami, B. (2023). Quantizing Yolov7: A comprehensive study. *2023 28th International Computer Conference, Computer Society of Iran (CSICC)*, 01-05. <https://doi.org/10.1109/csicc58665.2023.10105310>
- Cui, H., Radosavljevic, V., Chou, F.-C., Lin, T.-H., Nguyen, T., Huang, T.-K., Schneider, J., & Djuric, N. (2019). Multimodal trajectory predictions for autonomous driving using deep convolutional networks. *2019 International Conference on Robotics and Automation (ICRA)*, 2090-2096. <https://doi.org/10.1109/icra.2019.8793868>
- Dasgupta, K., Das, A., Das, S., Bhattacharya, U., & Yogamani, S. (2022). Spatio-contextual deep network-based multimodal pedestrian detection for autonomous driving. *IEEE Transactions on Intelligent Transportation Systems*, 23(9), 15940-15950. <https://doi.org/10.1109/tits.2022.3146575>
- Gao, J., Qiu, Y., & Chen, Z. (2025). Systemic integration of EV and autonomous driving technologies: A study of China's intelligent mobility transition. *World Electric Vehicle Journal*, 16(10), 574. <https://doi.org/10.3390/wevj16100574>
- Geiger, A., Lenz, P., & Urtasun, R. (2012). Are we ready for autonomous driving? The Kitty Vision Benchmark Suite. *2012 IEEE Conference on Computer Vision and Pattern Recognition*, 3354-3361. <https://doi.org/10.1109/cvpr.2012.6248074>
- Guindel, C. (2019). *eval_kitti* [Computer software]. GitHub. https://github.com/cguindel/eval_kitti
- Hui, J. H. (2018, March 7). *Map (mean average precision) for object detection*. Medium. <https://jonathan-hui.medium.com/map-mean-average-precision-for-object-detection-45c121a31173>

Levinson, J., Askeland, J., Becker, J., Dolson, J., Held, D., Kammel, S., Kolter, J. Z., Langer, D., Pink, O., Pratt, V., Sokolsky, M., Stanek, G., Stavens, D., Teichman, A., Werling, M., & Thrun, S. (2011). Towards fully autonomous driving: Systems and algorithms. *2011 IEEE Intelligent Vehicles Symposium (IV)*, 163-168. <https://doi.org/10.1109/ivs.2011.5940562>

Shah, D. (2023, May 30). *Intersection over union (IOU): Definition, calculation, code*. V7. <https://www.v7labs.com/blog/intersection-over-union-guide>

Sharma, V. (2021, July 7). *Kitti coordinate transformations*. Medium. <https://towardsdatascience.com/kitti-coordinate-transformations-125094cd42fb/>

Shi, P., Yang, L., Dong, X., Qi, H., & Yang, A. (2025). Research progress on multi-modal fusion object detection algorithms for autonomous driving: A review. *Computers, Materials & Continua*, 83(3), 3877-3917. <https://doi.org/10.32604/cmc.2025.063205>

Waymo LLC. (2019). *Waymo Open Dataset* [Data set]. <https://waymo.com/open/>

Yurtsever, E., Lambert, J., Carballo, A., & Takeda, K. (2020). A survey of autonomous driving: Common practices and emerging technologies. *IEEE Access*, 8, 58443-58469. <https://doi.org/10.1109/access.2020.2983149>